

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.	BL1 – Remember	5
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time.	BL2 – Understand	5

	Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.		
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: - courses, including number, title, credits, syllabus, and prerequisites; - course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; - students, including student-id, name, and program; - instructors, including identification number, name, department, and title. - Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5
4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. • A department controls a number of PROJECT. Projects have a unique name, number and a single location. • Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work	BL2 – Understand	5

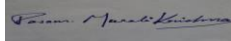
	for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).		
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5
8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I PASAM.MURALI KRISHNA(192525178) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. UPS Product Tracking Information System

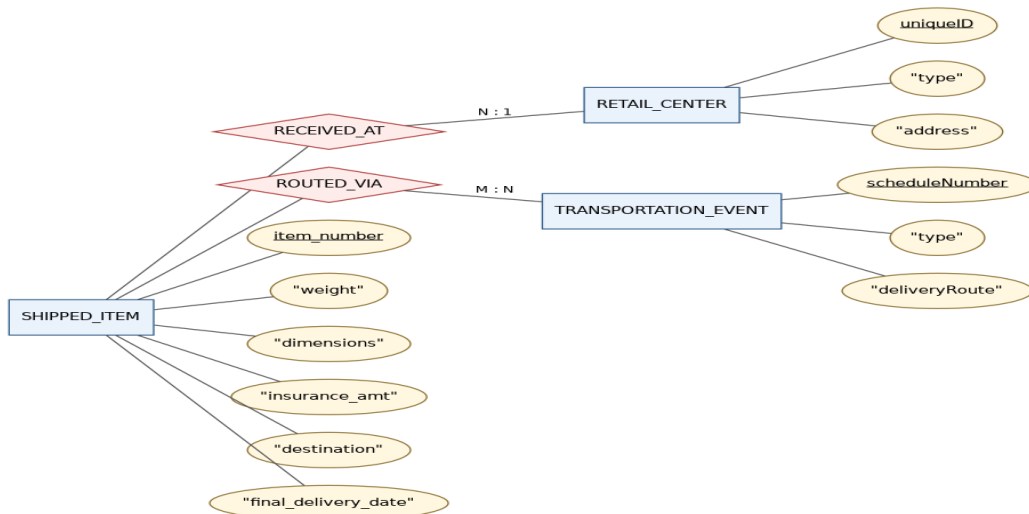
Entities

- SHIPPED_ITEM (item_number PK, weight, dimensions, insurance_amount, destination, final_delivery_date)
- RETAIL_CENTER (uniqueID PK, type, address)
- TRANSPORTATION_EVENT (scheduleNumber PK, type, deliveryRoute)

Relationships

- RECEIVED_AT — SHIPPED_ITEM (N) : RETAIL_CENTER (1)
- ROUTED_VIA — SHIPPED_ITEM (M) : TRANSPORTATION_EVENT (N)

E-R Diagram



Q2. Course Management System (Exam Scheduling)

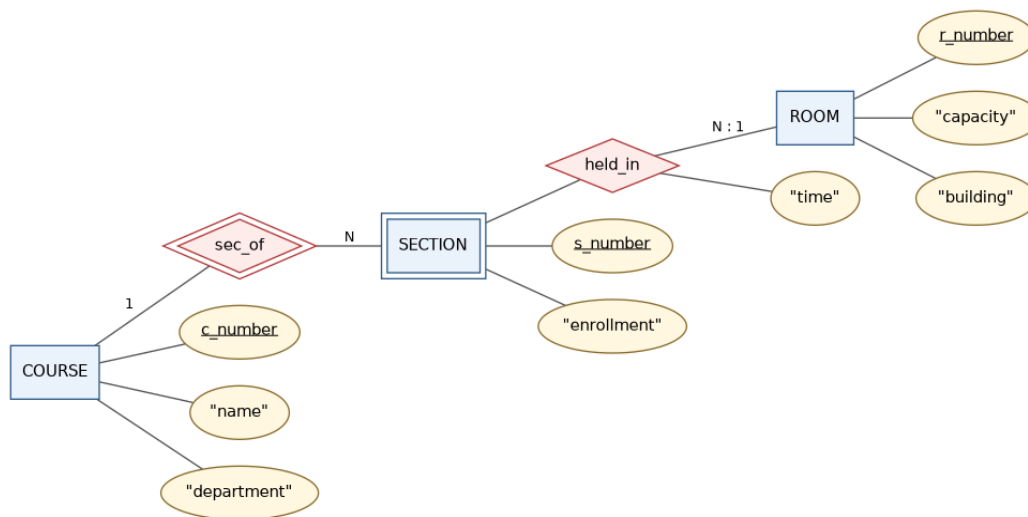
Entities

- course (c-number PK, name, department)
- section — weak entity (s-number partial key, enrollment), dependent on course
- room (r-number PK, capacity, building)

Relationships

- sec_of (identifying) — course (1) : section (N)
- held_in — section (N) : room (1), attribute time

E-R Diagram



Q3. University Registrar's Office Management System

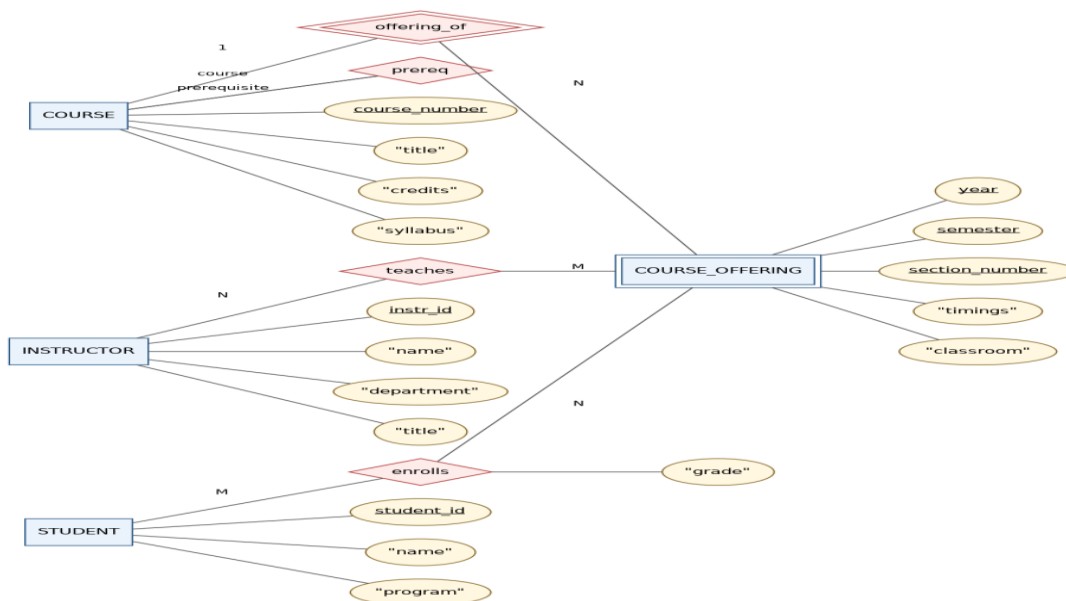
Entities

- course (course_number PK, title, credits, syllabus)
- course_offering — weak entity (year, semester, section_number partial key; timings, classroom)
- student (student_id PK, name, program)
- instructor (instructor_id PK, name, department, title)

Relationships

- offering_of (identifying) — course (1) : course_offering (N)
- teaches — instructor (N) : course_offering (M)
- enrolls — student (M) : course_offering (N), attribute grade
- prereq — recursive relationship on course

E-R Diagram



Q4. Hospital Management System

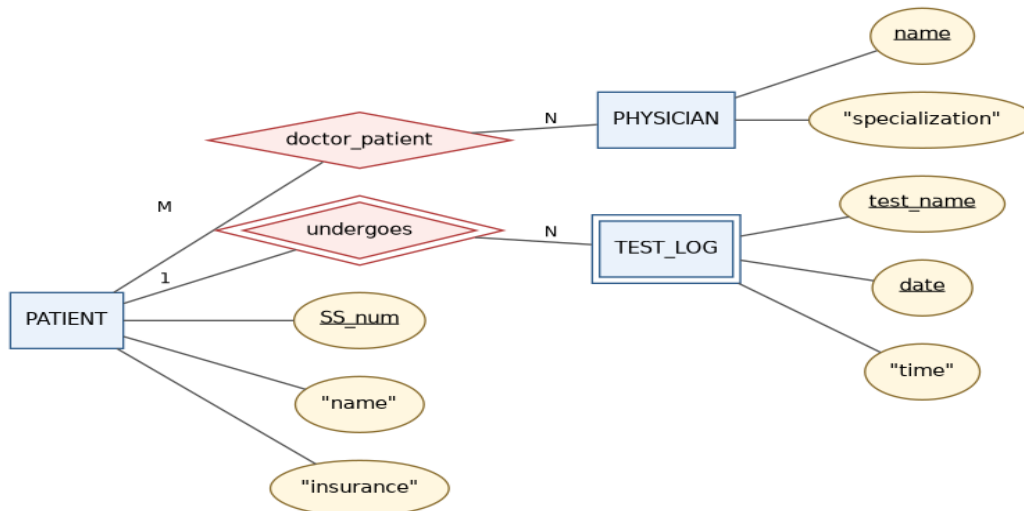
Entities

- PATIENT (SS# PK, name, insurance)
- PHYSICIAN (name PK, specialization)
- TEST_LOG — weak entity dependent on PATIENT (test-name, date partial key; time)

Relationships

- doctor_patient — PATIENT (M) : PHYSICIAN (N)
- undergoes (identifying) — PATIENT (1) : TEST_LOG (N)

E-R Diagram



Relational Tables

Table	Attributes	Key / Notes
Patient	SS#, name, insurance	PK: SS#
Physician	name, specialization	PK: name
Test-log	SS#, test-name, date, time	PK: (SS#, test-name, date); SS# → FK
Doctor-patient	physician-name, SS#	Composite PK; both FKs
Patient-history	SS#, test-name, date	FK → Patient

Q5. Company Project Management System

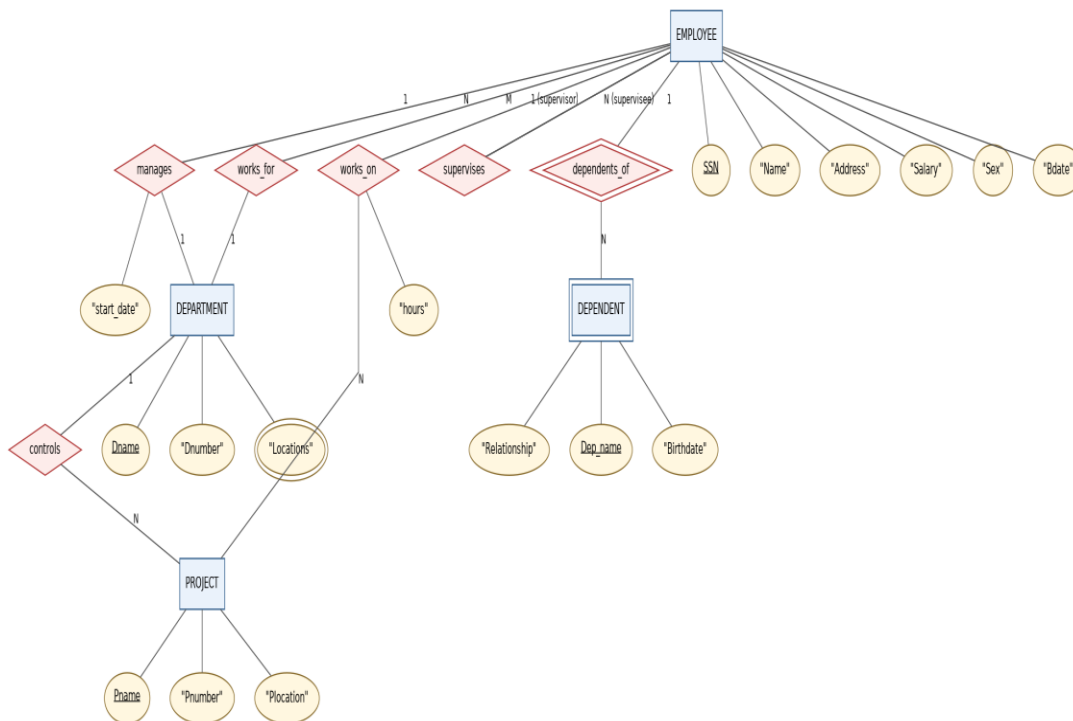
Entities

- DEPARTMENT (Dname PK, Dnumber, Locations — multivalued)
- PROJECT (Pname PK, Pnumber, Plocation)
- EMPLOYEE (SSN PK, Name, Address, Salary, Sex, Bdate)
- DEPENDENT — weak entity on EMPLOYEE (Dependent_name partial key; Birthdate, Relationship)

Relationships

- manages — EMPLOYEE (1) : DEPARTMENT (1), attribute start_date
- controls — DEPARTMENT (1) : PROJECT (N)
- works_for — EMPLOYEE (N) : DEPARTMENT (1)
- works_on — EMPLOYEE (M) : PROJECT (N), attribute Hours
- supervision — recursive on EMPLOYEE (1 : N)
- dependents_of (identifying) — EMPLOYEE (1) : DEPENDENT (N)

E-R Diagram



Q6. Schema, Instance, and Types of Schema

Schema vs Instance

- **Schema** — the overall design/structure of a database (tables, attributes, data types, constraints); rarely changes. E.g. STUDENT(roll_no, name, department, year).
- **Instance** — the actual data stored at a given moment; changes with every insert/update/delete. E.g. {(101,"Arjun","CSE",2)}.

Logical, Physical, and View Schema

Schema Type	Describes	Example
Physical	Low-level storage — file organization, indexing.	B+-tree indexed file on disk
Logical	Overall structure — tables, columns, constraints.	STUDENT(roll_no PK, name, dept, year)
View (external)	Subset a user/app is allowed to see.	App view: (roll_no, name, dept) only

Q7. Bank Management System

Bloom's Level: BL2 – Understand | Marks: 5

Entities

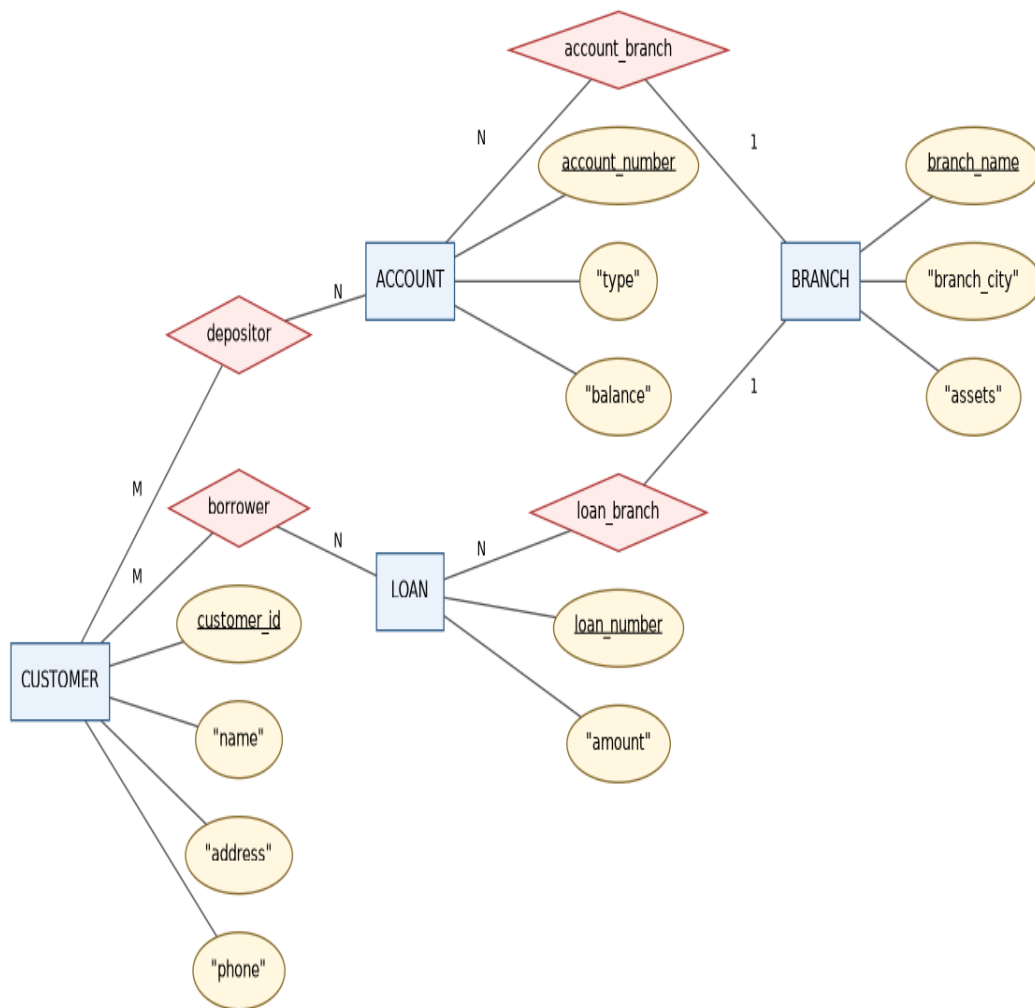
- **CUSTOMER** (customer_id PK, name, address, phone)

- ACCOUNT (account_number PK, type, balance)
- LOAN (loan_number PK, amount)
- BRANCH (branch_name PK, branch_city, assets)

Relationships

- depositor — CUSTOMER (M) : ACCOUNT (N)
- borrower — CUSTOMER (M) : LOAN (N)
- account_branch — ACCOUNT (N) : BRANCH (1)
- loan_branch — LOAN (N) : BRANCH (1)

E-R Diagram



Q8. Extended E-R (EER) Model

The EER model extends the basic E-R model with three concepts to express type hierarchies and higher-level relationships:

Generalization

Bottom-up: combining entity types with common attributes into a general superclass. E.g. CAR and TRUCK → VEHICLE.

Specialization

Top-down: dividing a superclass into subclasses based on distinguishing features (subclasses inherit superclass attributes). E.g. EMPLOYEE → SECRETARY, ENGINEER, MANAGER.

- Total vs partial — every entity must (total) or need not (partial) belong to a subclass.
- Disjoint vs overlapping — entity belongs to only one (d) or possibly several (o) subclasses.

Aggregation

Treats a relationship set as a higher-level entity so it can participate in another relationship. E.g. WORKS_ON(EMPLOYEE, PROJECT) aggregated to relate with EQUIPMENT via USES.

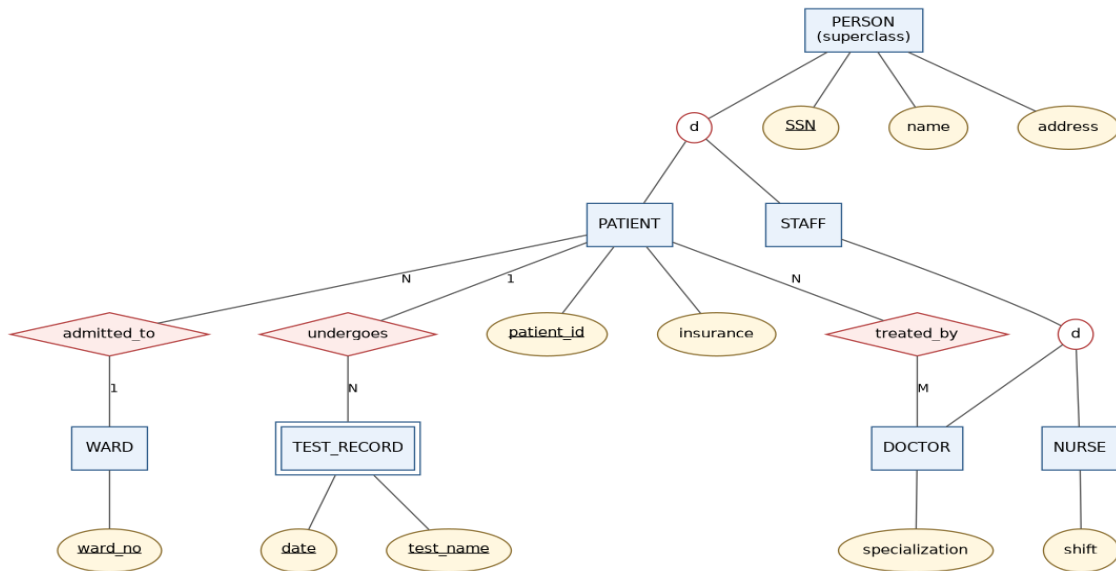
Q9. EER Diagram for a Hospital Management System

Design Overview

- PERSON — superclass (SSN, name, address); specialized (disjoint) into PATIENT and STAFF

- STAFF further specialized (disjoint) into DOCTOR (specialization) and NURSE (shift)
- PATIENT adds patient_id, insurance
- TEST_RECORD — weak entity on PATIENT (test_name, date partial key)
- Relationships: undergoes (identifying), treated_by, admitted_to

EER Diagram



Q10. Role of the Database Administrator (DBA)

The DBA is the person/team with central control over the data and programs accessing it, responsible for the DBMS's overall management, security, and performance.

Key Responsibilities

- Schema definition (logical, physical, view)
- Storage structure and access-method definition
- Schema and storage reorganization
- Granting authorization (GRANT/REVOKE)
- Routine maintenance and backup
- Ensuring data integrity and security
- Performance monitoring and tuning
- Backup and recovery planning